

STRELKA, Russian Federation

WMO: 292740

Lat: 58.091N

Lon: 93.162E

Elev: 294

StdP: 14.54

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.2	(c) -34.3	(d) -45.3	(e) 0.4	(f) -40.6	(g) -38.3	(h) 0.6	(i) -32.6	(j) 16.1	(k) 6.3	(l) 14.5	(m) 3.3	(n) 1.5	(o) 50	(p) 0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	84.1	67.2	80.9	65.5	77.7	64.1	69.4	79.6	67.7	77.4	66.0	74.6	5.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	96.8	73.9	64.2	90.8	72.3	62.4	85.2	70.4	33.7	79.7	32.4	77.7	31.0	74.6	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.8	(b) 13.7	(c) 12.0	(e) -44.5	(f) 90.0	(g) 5.8	(h) 2.4	(i) -48.7	(j) 91.7	(k) -52.1	(l) 93.1	(m) -55.3	(n) 94.5	(o) -59.6	(p) 96.2		
(4) 15.8	(5) 13.7	(6) 12.0	(7) -43.3	(8) 72.5	(9) 6.6	(10) 2.6	(11) -48.1	(12) 74.4	(13) -52.0	(14) 75.9	(15) -55.7	(16) 77.3	(17) -60.5	(18) 79.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.9	-6.6	2.1	18.2	35.0	47.5	61.2	66.0	61.2	47.0	32.8	11.7	4.6	
	DBStd	26.92	17.60	14.69	13.38	7.74	8.86	7.90	5.64	6.23	7.85	9.04	15.43	16.54	
	HDD50	7941	1754	1340	986	451	157	10	0	3	148	535	1149	1408	
	HDD65	12254	2219	1760	1451	900	547	160	56	150	539	1000	1599	1873	
	CDD50	1333	0	0	0	2	79	345	497	350	60	0	0	0	
	CDD65	169	0	0	0	0	3	46	88	32	0	0	0	0	
	CDH74	1574	0	0	0	1	61	531	721	255	5	0	0	0	
	CDH80	357	0	0	0	0	13	129	171	44	0	0	0	0	
Wind	WSAvg	5.4	5.2	5.1	5.5	6.3	5.9	4.4	4.0	4.2	5.2	6.3	6.6	6.4	
Precipitation	PrecAvg	18.7	1.0	0.8	0.7	1.1	1.7	2.2	2.0	2.4	2.0	1.7	1.6	1.5	
	PrecMax	28.1	2.0	1.2	1.6	1.5	3.6	4.3	6.5	4.6	4.2	3.2	2.4	2.4	
	PrecMin	14.4	0.1	0.3	0.0	0.2	0.2	0.1	0.3	1.0	0.4	0.6	0.7	0.9	
	PrecStd	3.7	0.4	0.3	0.4	0.3	1.0	1.1	1.5	1.0	0.9	0.7	0.4	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.5	33.5	47.1	63.3	81.7	88.0	88.3	84.6	73.7	56.4	43.6	33.6	
		MCWB	28.3	29.9	38.0	48.2	60.6	69.2	69.5	69.5	60.0	47.1	40.7	30.9	
	2%	DB	25.9	30.0	42.0	56.8	73.5	83.6	84.7	80.3	67.5	51.7	38.6	30.9	
		MCWB	23.9	27.2	34.9	44.9	56.7	66.0	67.8	66.1	57.0	44.0	35.5	28.9	
	5%	DB	22.6	26.9	38.6	51.3	68.0	80.3	81.2	76.6	63.0	47.6	35.4	27.8	
		MCWB	21.1	24.8	32.8	41.5	53.4	64.3	66.7	64.6	54.8	42.7	32.8	26.1	
	10%	DB	18.6	23.1	35.6	46.9	62.9	76.4	78.1	72.7	59.2	44.0	32.1	24.7	
		MCWB	17.2	21.2	31.3	39.1	51.2	62.6	64.8	63.1	53.2	40.4	30.3	23.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.4	30.0	38.9	49.7	62.5	71.5	72.7	70.3	61.3	48.4	41.1	31.5
			MCDB	30.4	32.4	44.7	61.5	77.7	84.0	83.7	81.2	70.5	54.1	43.8	32.8
2%		WB	24.2	27.5	35.6	45.4	58.2	68.3	70.0	68.1	58.3	45.1	35.8	29.2	
		MCDB	25.6	29.7	40.2	54.8	70.2	79.5	80.0	76.8	65.0	50.0	37.7	30.8	
5%		WB	21.1	24.8	33.4	42.5	55.2	66.5	68.3	66.2	56.0	42.8	33.1	26.3	
		MCDB	22.5	27.1	37.8	50.0	65.9	77.4	78.1	73.6	61.8	46.5	35.2	27.8	
10%		WB	17.4	21.2	31.5	39.5	52.5	64.0	66.6	64.1	53.5	40.6	30.2	23.2	
		MCDB	18.7	23.0	35.9	46.2	61.6	73.4	75.3	70.9	58.8	43.9	32.4	24.7	
Mean Daily Temperature Range	5% DB	MDBR	13.2	17.5	20.0	16.3	18.5	19.7	18.9	16.5	13.1	9.2	10.1	11.2	
		MCDBR	14.2	17.3	18.8	23.9	28.2	27.1	24.7	22.7	20.5	14.0	10.7	12.4	
		MCWBR	13.4	15.3	13.8	14.7	14.2	11.7	10.4	11.0	11.3	9.3	9.6	11.7	
	5% WB	MCDBR	14.4	16.2	16.9	22.1	25.7	24.3	21.1	19.4	17.8	12.8	10.4	12.1	
		MCWBR	13.6	14.4	13.1	14.4	14.1	11.7	10.2	10.6	10.8	9.4	9.5	11.6	
Clear-Sky Solar Irradiance	taub	0.289	0.320	0.357	0.389	0.404	0.419	0.448	0.414	0.368	0.344	0.304	0.278		
	taud	1.957	1.963	1.945	2.052	2.225	2.277	2.200	2.274	2.359	2.233	2.006	1.921		
	Ebn at Noon	156	211	240	256	262	260	248	248	243	207	148	117		
	Edh at Noon	19	31	42	44	40	39	41	36	28	24	18	15		
All-Sky Solar Radiation	RadAvg	193	489	908	1371	1583	1870	1716	1326	783	376	183	107		
	RadStd	29	43	74	98	161	187	121	107	114	39	35	23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page